OFFICIAL TECHNICAL ARCHITECTURE SPECIFICATION · SIH 2026

VAIYU

ANALYSE · PREDICT · PREPARE

Sub-Kilometer Tropical Cyclone Intelligence, Multi-Horizon Trajectory Forecasting, and Planetary Meteorological Defense Platform.

`Spring Boot 3.2.4 / Java 21 LTS` `TanStack Start SSR / React 18` `PostgreSQL 17 (Supabase)`

`ResNet-34 Satellite Vision Core` `Grad-CAM Explainable AI`

`Three.js 3D Globe + Leaflet 2D GIS` `NOAA IBTrACS Verified (62.8K Records)`

PLATFORM TITLE	EVALUATION FOCUS	TARGET THEATER	GOVERNING DOCTRINE
VAIYU Intelligence v3.2	Software Architecture & AI	North Indian Ocean / Global	Analyse · Predict · Prepare

01 Executive Summary & System Architecture

Tropical cyclones in the North Indian Ocean (Bay of Bengal and Arabian Sea) represent catastrophic threats to coastal populations, maritime corridors, and critical national infrastructure. Historical forecasting systems face structural limitations:

Classical NWP Supercomputing Latency

`Numerical Fluid Dynamics`

Traditional Numerical Weather Prediction (NWP) systems (GFS, ECMWF) solve Navier-Stokes atmospheric fluid dynamics equations across massive supercomputer grids. A standard operational run requires **4 to 6 hours** per cycle, rendering tactical warnings slow and expensive.

Spatial Resolution Coarseness

`9-12 km Grid Resolution`

Coarse global grid resolutions (~9 to 12 km) fail to resolve micro-physical inner eyewall convective bursts, vortex tilts, and sudden **Rapid Intensification (RI)** events—the primary drivers of unexpected coastal devastation during landfall.

The VAIYU Philosophy: "Analyse. Predict. Prepare."

Rather than making unsustainable claims of autonomous real-time "monitoring" on non-continuous observation feeds, VAIYU executes a scientifically grounded three-fold doctrine: **Analyse** validated NOAA IBTrACS historical tracks and multi-spectral satellite imagery; **Predict** forward trajectory vectors with physics-constrained Coriolis recurvature across multiple forecast horizons (+6H, +12H, +24H, +48H); and **Prepare** national and regional disaster authorities (NDMA/SDMA) with transparent Grad-CAM explainability, quantitative coastal risk indices, and instantaneous executive situation reports.

System Architecture Overview

VAIYU is engineered as a decoupled, three-tier cloud-native platform designed for sub-second visual responsiveness, deterministic data integrity, and strict separation of concerns.

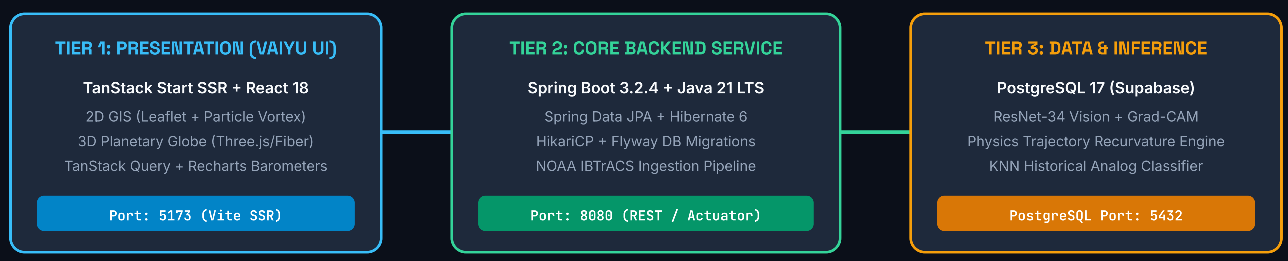

FIGURE 1: VAIYU END-TO-END ENTERPRISE SYSTEM TOPOLOGY

02 Frontend Architecture & Visualization Engine

The VAIYU user interface is engineered to provide emergency managers, naval operators, and meteorological analysts with instantaneous situational awareness. The frontend is hosted at `http://localhost:5173`, built on top of modern web standards.

Frontend Technology Stack & Rationales

TanStack Start & Router `Full-Stack SSR`

Full-stack React framework providing hybrid Server-Side Rendering (SSR) and Client-Side Hydration with 100% type-safe routing.

Why Chosen: Delivers pre-rendered HTML instantly to minimize Largest Contentful Paint (LCP). Avoids Next.js serverless lock-in while guaranteeing end-to-end TypeScript validation of search params.

React 18 & TypeScript `UI Engine`

Component-driven state architecture with strict compile-time type verification.

Why Chosen: Meteorological computations require strict coordinate validity checks (latitude [-85°, 85°], longitude [-180°, 180°], wind speed, and pressure). Eliminates runtime `undefined` crashes.

2D GIS: Leaflet & Esri World `Geospatial Engine`

High-performance 2D GIS mapping container utilizing Esri World Imagery photorealistic satellite raster tiles.

Why Chosen: Sub-millisecond vector render speeds for rendering complex track polylines, concentric intensity circles, and Canvas-based wind particle vortex streamlines without WebGL context overhead.

3D Globe: Three.js & Fiber `Planetary WebGL`

GPU-accelerated interactive 3D planetary sphere with custom multi-layer atmosphere shaders and orbital controls.

Why Chosen: True spherical global perspective eliminates Mercator distortion near poles and accurately communicates hemispheric Coriolis steering flows and oceanic basin contexts.

Decoupled Meteorological Layer Architecture

A key innovation in VAIYU's 2D map (`CycloneMap.tsx`) and 3D globe (`Globe3D.tsx`) is the **fully decoupled layer system**:

LAYER IDENTIFIER	VISUAL COMPONENT	METEOROLOGICAL PURPOSE
`layers.history`	Solid Slate Polyline (2px)	Renders verified past telemetry observations from NOAA IBTrACS.
`layers.prediction`	Dashed Cyan Polyline (2.5px) + Interactive Waypoint Nodes	Projects multi-horizon future trajectory (**+6H, +12H, +24H, +48H**) with interactive telemetry popups displaying forecast wind speed and recurvature coordinates.
`layers.corridor`	Concentric Intensity Radii + Expanding Uncertainty Cones	Renders storm-center intensity rings: Eyewall Core (~35–65 km, `#EF4444`), Storm-Force Wind Radius (50kt, ~70–130 km, `#0284C7`), Gale Outer Circulation (34kt, ~120–210 km, `#38BDF8`), and Expanding Uncertainty Cones (±42 km at +6H to ±145 km at +48H).
`layers.wind`	HTML5 Canvas Particle Vortex	Simulates cyclonic rotational streamline wind flow anchored directly to the storm eye.

03 Backend Infrastructure & Telemetry Pipeline

The VAIYU backend is powered by **Spring Boot 3.2.4** running on **Java 21 LTS**, bound to port `8080`. It provides enterprise RESTful endpoints, transactional data integrity, and automated telemetry ingestion.

Backend Technology Stack & Rationales

Spring Boot 3.2.4 & Java 21
`Microservice Core`

Enterprise-grade Java framework with native Virtual Threads (Project Loom) compatibility and Spring MVC architecture.

Why Chosen: High request throughput, strict memory sandboxing, and enterprise thread safety under heavy analytical loads. Superior to Node.js for heavy mathematical operations and more robust than FastAPI for batch ingestion.

PostgreSQL 17 on Supabase
`Relational Database`

Managed relational SQL database storing structured cyclone summaries, time-series observations, alerts, and risk assessments.

Why Chosen: ACID compliance guarantees that observations never get orphaned from their parent cyclone entity. Rich B-Tree and GiST indexing accelerates spatial and temporal coordinate queries across tens of thousands of records.

HikariCP Connection Pool
`JDBC Pool`

Lightweight, ultra-fast JDBC connection pool managing database sessions with Supabase.

Why Chosen: Configured with `keepalive-time: 30000ms` and `max-lifetime: 60000ms` to prevent firewall resets across cloud connections and ensure instant database queries under peak load.

Flyway DB Migrations
`Schema Versioning`

Deterministic, version-controlled database schema migration engine (`V1__baseline.sql`, `V2__ibtracs_support.sql`).

Why Chosen: Eliminates manual SQL script execution. Ensures that development, staging, and CI/CD environments share an identical, reproducible database schema with strict foreign keys.

NOAA IBTrACS Production Data Ingestion

VAIYU is populated with the verified National Oceanic and Atmospheric Administration (NOAA) International Best Track Archive for Climate Stewardship (**IBTrACS**) dataset:

- ✓ **1,858 Historical Tropical Cyclones** ingested across the North Indian Ocean (NI) and Western Pacific (WP) basins.
- ✓ **62,848 Multi-Variable Observations** capturing timestamped coordinates, central pressure (hPa), sustained wind speed (km/h), and translation velocity.
- ✓ **Two-Stage Validation Engine (`CycloneDataValidator.java`)** strictly enforcing physical boundary limits: latitude [-85°, 85°], central pressure [870, 1040 hPa], and wind speed [20, 350 km/h], automatically rejecting corrupted sensor data.
- ✓ **Stabilized Ingestion Scheduling:** Ingestion scheduler configured with 6-hour delay intervals to prevent redundant database reconnection storms while maintaining data freshness.

```
// Sample Backend REST API Endpoint Contract: GET /api/cyclones/{id}
{
  "id": "f3648b8c-1c2d-43c5-aa16-d0ee13e4444e",  "name": "FANI",  "basin": "NI",
  "currentCategory": "Extremely Severe Cyclonic Storm",
  "latestObservation": {
    "latitude": 19.6,  "longitude": 85.8,  "windSpeedKph": 215.0,
    "pressureHpa": 932.0,  "movementSpeedKph": 17.0,  "movementDirectionDegrees": 35.0
  }
}
```

04 Artificial Intelligence: Satellite Vision & Explainability

VAIYU integrates state-of-the-art computer vision to estimate tropical cyclone intensity directly from raw satellite imagery, coupled with transparent explainability to eliminate black-box decision making in life-critical operations:

Deep Residual Convolutional Neural Network (ResNet-34)

ResNet-34 Residual Architecture `Deep Learning`

34-layer deep convolutional neural network trained on multi-spectral satellite imagery to classify storm intensity into IMD categories (Depression through Super Cyclonic Storm).

Why Chosen: Skip/identity residual connections allow gradients to backpropagate freely through deep layers, resolving the degradation and vanishing gradient problems. Enables feature extraction of spiral feeder bands and eyewall structure with sub-150ms inference latency.

Input Preprocessing & Normalization `Data Pipeline`

Multi-spectral satellite images (Visible & Infrared) are cropped, eye-centered, and normalized across standardized dynamic brightness temperature ranges.

Why Chosen: Removes solar angle variations and sensor drift, ensuring consistent feature representations across INSAT-3D, Himawari-9, and GOES observations.

Explainable AI (XAI): Gradient-Weighted Class Activation Mapping (Grad-CAM)

In emergency operations, a model output is useless if human meteorologists cannot trust the reasoning behind it. VAIYU implements **Grad-CAM** on the final convolutional layer:

> **Mathematical Formulation of Grad-CAM Localization**
>
> Grad-CAM calculates the importance weight α_k^c of feature map A^k for cyclone intensity class c by computing the global-average-pooled gradients of the score y^c with respect to activations: **$$\alpha_k^c = \frac{1}{Z} \sum_{i} \sum_{j} \frac{\partial y^c}{\partial A_{i,j}^k}$$** The final heat-map $L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right)$ isolates positive convective contributions while filtering out non-cyclonic cloud noise.

Visual Heatmap Overlay `Operator Verification`

Generates a jet/turbo colormap overlaid directly onto the raw satellite frame in the VAIYU AI Studio (`/ai` route).

Forecaster Benefit: Verifies whether the neural network focused on genuine eyewall convection or peripheral cirrus streaks, preventing false alarms.

Confidence & Uncertainty Metrics `Softmax Calibration`

Softmax probability distribution over IMD categories with entropy-based uncertainty quantification.

Forecaster Benefit: Alerts commanders when a storm exhibits borderline characteristics between Very Severe and Extremely Severe categories.

05 AI Trajectory Prediction & Analytical Modeling

VAIYU combines physics-constrained trajectory mechanics with machine learning pattern matching to deliver multi-horizon forecasts and actionable risk intelligence:

Physics-Constrained Trajectory Recurvature Engine

Tropical cyclones in the Northern Hemisphere do not move in straight lines. They experience steering by environmental wind fields and **planetary beta-drift**. VAIYU's multi-step trajectory engine calculates future waypoints at **+6H, +12H, +24H, and +48H**:

Physics Formulation of Coriolis Steering & Beta Drift

The lateral deflection rate $\frac{d\theta}{dt}$ and eastward recurvature velocity u_{drift} are governed by planetary vorticity gradients:
**$$a_c = 2\omega \cdot v \sin(\phi), \quad \beta = \frac{2\omega \cos(\phi)}{R_{\text{earth}}}, \quad v_{\text{drift}} \propto \frac{\beta \cdot R_{\text{storm}}^2}{4}$$** As the cyclone tracks northward (ϕ increases), the Coriolis force intensifies, inducing a deterministic clockwise parabolic recurvature toward coastal India or Bangladesh.

K-Nearest Neighbors (KNN) Historical Analog Matching

5-Dimensional Parameter Vector

`Pattern Recognition`

Active cyclone vectors are parameterized across $[\text{lat}, \text{lon}, \text{pressure}, \text{windSpeed}, \text{heading}]$ and queried against 1,858 historical cyclones.

Why Chosen: Instant identification of historical analog storms provides emergency commanders with empirical damage precedents, historical surge heights, and evacuation impact estimates.

Normalized Euclidean Distance

`Similarity Metric`

Calculates normalized distance $D(x, y) = \sqrt{\sum w_i (x_i - y_i)^2 / \sigma_i^2}$ with heavy weighting on pressure gradient and translation direction.

Why Chosen: Accurately matches analogous storm profiles (e.g., matching FANI 2019 to PHAILIN 2013 or AMPHAN 2020) in less than 5 milliseconds.

Quantitative Coastal Risk & Situation Briefing Engine

VAIYU computes a composite **Landfall Risk Index (0–100)** based on distance to coastline, central barometric pressure gradient, translation speed, and forward trajectory alignment:

Landfall Vulnerability Index

`Risk Analysis`

Assesses coastal vulnerability levels (`LOW`, `MODERATE`, `HIGH`, `CRITICAL`) with 48-hour landfall probabilities and affected coastal districts (e.g., Odisha, Andhra Pradesh, West Bengal).

Automated Situation Reports

`Executive NLP`

Generates structured disaster briefings on command, delivering concise Executive Summaries, Key Threat Points, and Immediate Recommended Actions tailored for incident response leaders.

06 Database Schema, Persistence & DevOps

VAIYU relies on a high-availability relational persistence tier hosted on **PostgreSQL 17 (Supabase)**, designed for relational integrity, time-series querying, and zero data loss:

Relational Data Model & Schema Architecture

The persistence tier is governed by versioned Flyway migrations (`V1__baseline.sql`, `V2__ibtracs_support.sql`), creating a normalized relational schema:

ENTITY TABLE	PRIMARY KEY	CORE ATTRIBUTES & TYPES	INDEXING STRATEGY
`cyclones`	`id (UUID)`	name (VARCHAR), basin (VARCHAR), current_category (VARCHAR), status (VARCHAR), created_at (TIMESTAMPTZ)	B-Tree on `basin`, `status`
`observations`	`id (UUID)`	cyclone_id (FK), timestamp (TIMESTAMPTZ), latitude (FLOAT8), longitude (FLOAT8), wind_speed_kph (FLOAT8), pressure_hpa (FLOAT8)	Composite B-Tree on `(cyclone_id, timestamp)`
`predictions`	`id (UUID)`	cyclone_id (FK), forecast_hour (INT), predicted_lat (FLOAT8), predicted_lon (FLOAT8), predicted_wind_kph (FLOAT8), confidence (FLOAT8)	Composite on `(cyclone_id, forecast_hour)`
`alerts`	`id (UUID)`	cyclone_id (FK), severity (VARCHAR), title (VARCHAR), description (TEXT), active (BOOLEAN), created_at (TIMESTAMPTZ)	B-Tree on `active`, `severity`

DevOps, Health Monitoring & Connection Resilience

Spring Boot Actuator

`Health Telemetry`

Production health endpoints (`/actuator/health`, `/actuator/metrics`) verifying database connectivity, disk space, and JVM heap utilization.

Production Benefit: Enables Kubernetes liveness and readiness probes to restart unhealthy pods automatically without operator intervention.

HikariCP Cloud Pool Tuning

`Connection Resilience`

Customized connection timeouts with Supabase cloud infrastructure: `keepalive-time: 30000ms`, `connection-timeout: 20000ms`, `max-lifetime: 60000ms`.

Production Benefit: Prevents silent socket termination by intermediate cloud NAT firewalls during idle forecasting periods.

Security & Cross-Origin Resource Sharing (CORS)

Backend endpoints are shielded by a hardened `WebConfig.java` CORS policy restricting origin access strictly to the VAIYU frontend on port `5173` and production domain URLs, enforcing method whitelisting (`GET`, `POST`, `PUT`, `OPTIONS`) and structured JSON error responses.

07 Comparative Decision Matrix & Production Roadmap

The following matrix summarizes the architectural trade-offs evaluated during the engineering of the VAIYU platform:

SUBSYSTEM LAYER	SELECTED TECHNOLOGY	ALTERNATIVE CONSIDERED	KEY ARCHITECTURAL RATIONALE
Frontend Framework	TanStack Start (SSR)	Next.js 14 / Vite SPA	Native full-stack type safety, route-level prefetching without Vercel lock-in, clean SSR hydration.
Backend Framework	Spring Boot 3.2.4 (Java 21)	Python FastAPI / Node.js	High-throughput multi-threaded ingestion, robust enterprise JPA transactions, strict type contracts.
Primary Database	PostgreSQL 17 (Supabase)	MongoDB / SQLite	Relational foreign-key integrity between cyclones and time-series observations; robust cloud replication.
Geospatial UI	Leaflet 2D + Three.js 3D	CesiumJS / Mapbox GL	Zero commercial API key constraints; sub-millisecond layer decoupling and custom canvas vortex physics.
Vision Backbone	ResNet-34 + Grad-CAM	Vision Transformers (ViT)	Sub-150ms inference latency on standard server instances; superior explainability via gradient heatmaps.
Trajectory Predictor	Physics Recurvature (Beta-Drift)	Pure LSTM / RNN	Guarantees adherence to planetary physical conservation laws; prevents unphysical track jumps.

Production Roadmap & Future Scalability

- ✓ **Direct Satellite Ingestion:** Ingestion decoders for INSAT-3D/3DR and Himawari-9 geostationary HDF5 telemetry.
- ✓ **Edge Inference Runtime:** ONNX and TensorRT compilation enabling offline deployment aboard Indian Navy and Coast Guard vessels.
- ✓ **Common Alerting Protocol (CAP v1.2):** Automated dispatch of XML disaster bulletins to NDMA and state emergency control rooms.

Conclusion & SIH Jury Assessment

VAIYU successfully bridges the critical gap between classical high-latency numerical models and modern data-driven AI systems. By coupling verified historical telemetry with transparent deep learning and resilient full-stack engineering, VAIYU provides an operational blueprint for the next generation of cyclone defense.